

Title: Monitoring Chlorophyll Levels in Solina Reservoir Using Satellite Data

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Abstract:

Monitoring chlorophyll-a concentrations in inland water bodies is essential for assessing water quality, detecting algal blooms, and understanding ecological dynamics. This study focuses on the Solina Reservoir, one of Poland's largest artificial lakes, utilizing multi-source satellite observations and in-situ measurements to track chlorophyll-a levels over time. By integrating data from Sentinel-3 OLCI, Sentinel-2 MSI, and field measurements, this project aims to develop an efficient and accurate remote sensing-based approach to monitor water quality dynamics.

The methodology involves preprocessing Sentinel-3 OLCI data to extract chlorophyll indices using bio-optical models and band ratios. Additionally, Sentinel-2 MSI imagery, with its higher spatial resolution, is incorporated to refine chlorophyll-a concentration estimates and enhance spatial accuracy. Field measurements, including spectroradiometer readings and water sampling, serve as validation data to assess the accuracy of satellite-derived estimates. By applying time-series analysis, the study identifies seasonal and spatial variations in chlorophyll-a levels, providing insights into potential eutrophication trends and nutrient loading patterns.

The expected outcomes of this research include the generation of chlorophyll-a concentration maps, the analysis of temporal trends in water quality, and accuracy assessments of remote sensing techniques through comparison with in-situ measurements. The findings will contribute to improved monitoring of the Solina Reservoir, supporting water resource management and early detection of ecological changes. This project demonstrates the potential of satellite remote sensing in environmental monitoring and emphasizes its role in sustainable water management.